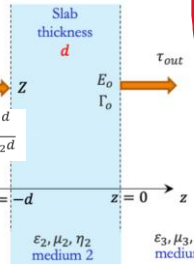
Single Slab (double-interface) Problem

Let's consider normal incidence only. All media are lossless.

The basic equations are $\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$ $\eta = \frac{E}{H}$ We can write $\Gamma_{in} = \frac{Z - \eta_1}{Z + \eta_1}$



The impedance at $z = -d$ is

$$Z = \frac{E^{tot}}{H^{tot}} = \frac{E^i + E^r}{H^i + H^r}$$

Now can evaluate the overall reflection coefficient.

$$= \frac{E_o\, e^{-j\beta_2(-d)} + \Gamma_o E_o\, e^{j\beta_2(-d)}}{\frac{E_o}{\eta_2} e^{-j\beta_2(-d)} - \Gamma_o \frac{E_o}{\eta_2} e^{j\beta_2(-d)}}$$

$$= \eta_2 \frac{1 + \Gamma_o\, e^{-j2\beta_2 d}}{1 - \Gamma_o\, e^{-j2\beta_2 d}}$$

$$\Gamma_{in} = \frac{Z - \eta_1}{Z + \eta_1} = \frac{\Gamma_{12} + \Gamma_{23}\, e^{-j2\beta_2 d}}{1 + \Gamma_{12}\Gamma_{23}\, e^{-j2\beta_2 d}}$$

$$\Gamma_{12} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \qquad \Gamma_{23} = \frac{\eta_3 - \eta_2}{\eta_3 + \eta_2}$$

The input impedance is

Substitute $\Gamma_o = \frac{\eta_3 - \eta_2}{\eta_3 + \eta_2}$

$$Z = \eta_2 \left\{ \frac{(\eta_3 + \eta_2) + (\eta_3 - \eta_2)\, e^{-j2\beta_2 d}}{(\eta_3 + \eta_2) - (\eta_3 - \eta_2)\, e^{-j2\beta_2 d}} \right\}$$

$$\tau_{out} = \frac{\tau_{12}\tau_{23}\, e^{-j\beta_2 d}}{1 + \Gamma_{12}\Gamma_{23}\, e^{-j2\beta_2 d}} \qquad \tau_{12} = 1 + \Gamma_{12} \qquad \tau_{23} = 1 + \Gamma_{23}$$

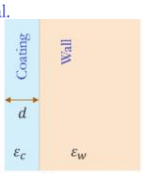
$$\approx \Gamma_{12} + \Gamma_{23}\, e^{-j2\beta_2 d}$$ for Γ_{12} and Γ_{23} samll

For a lossless slab Power flux in=Power flux out $\frac{1}{\eta_1}(1 - |\Gamma_{in}|^2) = \frac{1}{\eta_3}|\tau_{out}|^2 \qquad (1 - |\Gamma_{in}|^2) = \frac{\eta_1}{\eta_3}|\tau_{out}|^2$

Transmittance

2) A coating is needed on a building walls ($\varepsilon_{rw} = 4$) such that incident waves at 10 GHz from air are not reflected. Find the required coating thickness and material.

$\Gamma_{in} = 0$ $|\Gamma_{ac} + \Gamma_{cw}\, e^{-j2\beta_c d}| = 0$ $\Gamma_{ac} = \frac{\eta_c - \eta_a}{\eta_c + \eta_a}$ $\Gamma_{cw} = \frac{\eta_w - \eta_c}{\eta_w + \eta_c}$



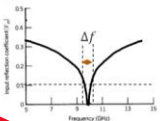
Look at $2\beta_c d = 2\frac{2\pi}{\lambda_c} d = \pi \frac{4}{\lambda_c} d$

If we take $d = \frac{\lambda_c}{4}$ then $2\beta_c d = \pi$

$e^{-j2\beta_c d} = e^{-j\pi} = -1$ $\left|\frac{\eta_c - \eta_a}{\eta_c + \eta_a} - \frac{\eta_w - \eta_c}{\eta_w + \eta_c}\right| = 0$

$$\left|\frac{(\eta_c - \eta_a)(\eta_w + \eta_c) - (\eta_w - \eta_c)(\eta_c + \eta_a)}{(\eta_c + \eta_a)(\eta_w + \eta_c)}\right| = \left|\frac{2(\eta_c^2 - \eta_a\eta_w)}{(\eta_c + \eta_a)(\eta_w + \eta_c)}\right| = 0 \quad \Rightarrow \quad 2|(\eta_c^2 - \eta_a\eta_w)| = 0$$

If we take $\eta_c = \sqrt{\eta_a\eta_w}$ $\Gamma_{in} = 0$ $d = \frac{\lambda_c}{4} = 0.53\ cm$



Also $\varepsilon_{rc} = \sqrt{\varepsilon_{ra}\varepsilon_{rw}}$ Quarter-wavelength Transformer Anti-reflection Coating

Application: Radome Design

1) A 10 GHz UPW is normally incident on a dielectric slab ($\varepsilon_r = 2.56$) of thickness d, with air on both sides of the slab. Find the required thickness for zero overall reflection at 10 GHz.

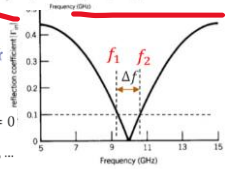
$\Gamma_{in} = 0$ $|\Gamma_{12} + \Gamma_{23}\, e^{-j2\beta_2 d}| = 0$ $\Gamma_{12} = -\Gamma_{23}$ $|\Gamma_{12}||1 - e^{-j2\beta_2 d}| = 0$

$e^{-j2\beta_2 d} = 1 = \cos(2\beta_2 d) - j\sin(2\beta_2 d)$ $2\beta_2 d = n(2\pi), n = 0,1,2,3,\ldots$

$d = n\left(\frac{\pi}{\beta_2}\right),\ 0, \frac{\pi}{\beta_2}, \frac{2\pi}{\beta_2}, \ldots$ $d = n\left(\frac{\lambda_2}{2}\right),\ 0, \frac{\lambda_2}{2}, \lambda_2, \frac{3\lambda_2}{2}, \ldots$

At 10 GHz, $\lambda_2 = \frac{c}{\sqrt{\varepsilon_{r2}} \cdot f} = 1.875\ cm$ $d = 0.9375\ cm$

How much reflection for 5 GHz $\Gamma_{in} = |\Gamma_{12}||1 - e^{-j2\beta_2 d}| = (0.231)|1 - e^{-j2\omega\sqrt{2.56}\frac{0.009375}{c}}| = 0.438$



Find the bandwidth for 10% field reflection

$(0.231)|1 - e^{-j2\beta_2 d}| = 0.1$

$f_1 = 9.3\ GHz$, $f_2 = 10.7\ GHz$, $BW = \Delta f = 1.4\ GHz$

Factional BW $\Delta f/f = 14\%$

1) Total Transmission. $|\Gamma| = 0$, Only TM polarization Incident Angle $\theta_i = \tan^{-1}(\sqrt{\varepsilon_{r2}/\varepsilon_{r1}})$ Brewster Angle θ_B

In this case $\theta_i + \theta_t = 90^o$ Air-Glass: $\theta_B = \tan^{-1}(1.52/1)$ $\theta_B = 56.7^o$

2) Total Reflection $|\Gamma| = 1$, Both TE and TM polarizations, for $\varepsilon_{r1} > \varepsilon_{r2}$ only. Incident Angle $\theta_i = \sin^{-1}(\sqrt{\varepsilon_{r2}/\varepsilon_{r1}})$

Critical Angle θ_c Glass-Air: $\theta_c = \sin^{-1}(1/1.52)$ $\theta_c = 41.1^o$ Total internal reflection (TIR)

Happens for any angle $\theta_i \geq \theta_c$

Single Interface: Normal Incidence

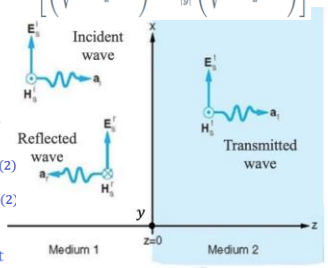
$$\sqrt{x + iy} = \pm\left[\left(\sqrt{\frac{\sqrt{x^2+y^2}+x}{2}}\right) + i\frac{y}{|y|}\left(\sqrt{\frac{\sqrt{x^2+y^2}-x}{2}}\right)\right]$$

Reflected Wave

$$\vec{E_r}(z) = \Gamma E_o\, e^{+j\beta_1 z}\, \vec{a}_x \qquad \vec{H_r}(z) = \frac{\Gamma E_o}{\eta_1} e^{+j\beta_1 z}\, (-\vec{a}_y)$$

Incident Wave

$$\vec{E_i}(z) = E_o\, e^{-j\beta_1 z}\, \vec{a}_x \qquad \vec{H_i}(z) = \frac{E_o}{\eta_1} e^{-j\beta_1 z}\, \vec{a}_y$$



Transmitted Wave

$$\vec{E_t}(z) = \tau E_o\, e^{-j\beta_2 z}\, \vec{a}_x \qquad \vec{H_t}(z) = \frac{\tau E_o}{\eta_2} e^{-j\beta_2 z}\, \vec{a}_y$$

$E_{tangential\,(1)} = E_{tangential\,(2)}$

$H_{tangential\,(1)} = H_{tangential\,(2)}$

Reflection coefficient $\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$ $\varepsilon_1, \mu_1, \sigma_1, \eta_1, \gamma_1$ $\varepsilon_2, \mu_2, \sigma_2, \eta_2, \gamma_2$

Apply boundary conditions

$E_{1_tan} = E_o(1 + \Gamma)$

$E_{2_tan} = E_o\tau$

$H_{1_tan} = \frac{E_o}{\eta_1}(1 - \Gamma)$

$H_{2_tan} = \frac{\tau E_o}{\eta_2}$

$\Gamma = 1 - \tau$

Transmission coefficient $\tau = \frac{2\eta_2}{\eta_1 + \eta_2}$

Power coefficients

$R = |\Gamma|^2$ Reflectance

$T = \frac{\eta_1}{\eta_2}|\tau|^2$ Transmittance

$R + T = 1$ Power Conservation

$$\vec{S}^i_{ave} = \frac{|E_o|^2}{2\eta_1}\vec{a}_k \qquad \vec{S}^r_{ave} = R\frac{|E_o|^2}{2\eta_1}(-\vec{a}_k) \qquad \vec{S}^t_{ave} = T\frac{|E_o|^2}{2\eta_1}\vec{a}_k$$

Remarks

1) Medium 1 perfect dielectric (η_1=real) and medium 2 perfect conductor (η_2=0) $\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = -1$ Total reflection

2) Medium 1 is a denser dielectric than medium 2 $\varepsilon_{r1} > \varepsilon_{r2}$ $\eta_2 > \eta_1$ $\Gamma > 0$ $\tau = \frac{2\eta_2}{\eta_1 + \eta_2} = 0$ Zero transmission

3) Medium 1 is a lighter dielectric than medium 2 $\varepsilon_{r1} < \varepsilon_{r2}$ $\eta_1 > \eta_2$ $\Gamma < 0$

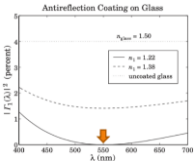
3) Design a ¼-wavelength antireflection coating for a glass substrate ($\varepsilon_{rg} = 2.25$) . The coating is supposed to work for mid-visible range ($\lambda \approx 550\ nm$) .



Using the quarter wavelength condition $\varepsilon_{rc} = \sqrt{\varepsilon_{ra}\varepsilon_{rg}} = \sqrt{2.25} = 1.5$ $d = \frac{\lambda_c}{4} = 112.7\ nm$

The solution is not *that* easy! Practically, the closest materials to the requirement are Cryolite ($\varepsilon_r = 1.82$) and Magnesium Fluoride ($\varepsilon_r = 1.91$, more durable for this task and more abundant).

$$\Gamma_{in} = \frac{\sqrt{\varepsilon_{ra}\varepsilon_{rg}} - \varepsilon_{rc}}{\sqrt{\varepsilon_{ra}\varepsilon_{rg}} + \varepsilon_{rc}} = -0.118 \qquad R = |\Gamma_{in}|^2 = 1.4\% \qquad \text{Without coating } R = 4\% \qquad d = \frac{\lambda_c}{4} = 99.6\ nm$$



Reflection coefficient $\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$

Transmission coefficient $\tau = \frac{2\eta_2}{\eta_1 + \eta_2}$

Example: Glass Reflection

Glass dielectric constant, $\varepsilon_r \approx 2.3$

$$\eta = \eta_o\sqrt{\frac{\mu_r}{\varepsilon_r}} = \eta_o\sqrt{\frac{1}{2.3}} = \frac{\eta_o}{1.52}$$

$$\Gamma = \frac{\frac{\eta_o}{1.52} - \eta_o}{\frac{\eta_o}{1.52} + \eta_o} = -0.2065$$

$R = 0.2065^2 = 0.0426$

$T = 1 - 0.0426 = 0.957$

96% of light power goes through!

Example: Seawater Reflection

Seawater dielectric constant, at 2.45 GHz and 30 °C is $\varepsilon_r = 75.9 - j8.6$

$$\eta_2 = \frac{\eta_o}{\sqrt{\varepsilon_r}} = \frac{377}{\sqrt{75.9 - j8.6}} = 43.1\angle 3.2^o\ \Omega$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\Gamma = \frac{43.1\angle 3.2^o - 377}{43.1\angle 3.2^o + 377}$$

$$\Gamma = \frac{-333.97 + j2.4}{420 + j2.4} = 0.8\angle -0.74^o$$

$R = 0.8^2 = 64\%$

$T = 36\%$

Lossy Media

(η_1 and η_2 complex qua

Electric Fields

$$\vec{E_i}(z) = E_o\, e^{-\alpha_1 z} e^{-j\beta_1 z}\, \vec{a}_x \qquad \vec{S}^i_{ave} = \frac{|E_o|^2}{2} e^{-2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)\vec{a}_z$$

$$\vec{E_r}(z) = \Gamma E_o\, e^{+\alpha_1 z} e^{+j\beta_1 z}\, \vec{a}_x \qquad \vec{S}^r_{ave} = |\Gamma|^2\frac{|E_o|^2}{2} e^{+2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)(-\vec{a}_z)$$

$$\vec{E_t}(z) = \tau E_o\, e^{-\alpha_2 z} e^{-j\beta_2 z}\, \vec{a}_x \qquad \vec{S}^t_{ave} = |\tau|^2\frac{|E_o|^2}{2} e^{-2\alpha_2 z}\, Re\left(\frac{\eta_1}{\eta_2}\right)\vec{a}_z$$

Example: Lossy dielectric reflection.

A plane wave in air at 1 GHz impinges normally on the surface of a lossy non-magnetic dielectric. The impedance of this dielectric at this frequency is $\eta = 214 + j82.3\ \Omega$. Find the reflection and transmission coefficients. Also, find the reflected and transmitted time-average power density vectors one meter away from the interface. Take $E_o = \sqrt{2}$ kV/m.

For medium 2, $\frac{\sigma}{\omega\varepsilon} = \tan 2\theta_\eta = \tan 42^o = 0.9$

$\varepsilon_r = 2.0$

$\alpha_2 = 12.3$ Np/m

$\Gamma = -0.2505 + j0.1739$

$\tau = 0.7495 + j0.1739$

Loss tangent ≈ 1

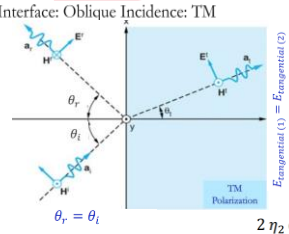
Loss cannot be neglected!

$\vec{S}^i_{ave} = \frac{|E_o|^2}{2} Re\left(\frac{1}{\eta_1}\right) e^{-2\alpha_1 z}\, \vec{a}_z = 2.65\, \vec{a}_z$ kW/m² Constant in air.

$\vec{S}^r_{ave} = |\Gamma|^2\frac{|E_o|^2}{2} Re\left(\frac{1}{\eta_1}\right) e^{+2\alpha_1 z}(-\vec{a}_z) = 0.246\,(-\vec{a}_z)$ kW/m²

$\vec{S}^t_{ave} = |\tau|^2\frac{|E_o|^2}{2} Re\left(\frac{\eta_1}{\eta_2}\right) e^{-2\alpha_2 z}\, \vec{a}_z = 0.5\, \vec{a}_z$ nW/m²

Single Interface: Oblique Incidence: TM



Incident Wave

$$\vec{E_i} = E_o\, e^{-j\beta_1(\cos\theta_i z + \sin\theta_i x)}\,(\cos\theta_i\, \vec{a}_x - \sin\theta_i\, \vec{a}_z)$$

$$\vec{H_i} = \frac{E_o}{\eta_1} e^{-j\beta_1(\cos\theta_i z + \sin\theta_i x)}\, \vec{a}_y$$

Reflected Wave

$$\vec{E_r} = \Gamma_{TM} E_o\, e^{-j\beta_1(\sin\theta_r x - \cos\theta_r z)}\,(\cos\theta_r\, \vec{a}_x + \sin\theta_r\, \vec{a}_z)$$

$$\vec{H_r} = \frac{\Gamma_{TM} E_o}{\eta_1} e^{-j\beta_1(\sin\theta_r x - \cos\theta_r z)}\,(-\vec{a}_y)$$

Transmitted Wave

$$\vec{E_t} = \tau_{TM} E_o\, e^{-j\beta_2(\cos\theta_t z + \sin\theta_t x)}(\cos\theta_t\, \vec{a}_x - \sin\theta_t\, \vec{a}_z)$$

$$\vec{H_t} = \frac{\tau_{TM} E_o}{\eta_2} e^{-j\beta_2(\cos\theta_t z + \sin\theta_t x)}\, \vec{a}_y$$

$\theta_r = \theta_i$

$\beta_1 \sin\theta_i = \beta_2 \sin\theta_t$

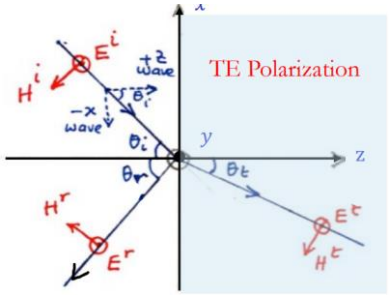
$\sqrt{\varepsilon_{r1}}\sin\theta_i = \sqrt{\varepsilon_{r2}}\sin\theta_t$

$$\tau_{TM} = \frac{2\,\eta_2\cos\theta_i}{\eta_1\cos\theta_i + \eta_2\cos\theta_t} \qquad R_{TM} = |\Gamma_{TM}|^2$$

$$\Gamma_{TM} = \frac{-\eta_1\cos\theta_i + \eta_2\cos\theta_t}{\eta_1\cos\theta_i + \eta_2\cos\theta_t} = \frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)} \qquad T_{TM} = \frac{\eta_1\cos\theta_t}{\eta_2\cos\theta_i}|\tau_{TM}|^2$$

$T_{TM} + R_{TM} = 1$

A single-frequency (0.1 GHz) plane wave is incident obliquely from air onto a planar interface with a non-magnetic medium having a dielectric constant of 2. The electric field of the incident wave has the following phasor form $\vec{E} = [\frac{1}{\sqrt{2}}\vec{a}_y]\, e^{-j\beta(z-\sqrt{2}x)/\sqrt{3}}$ V/m



From the $\vec{E}_i$ equation, the z incident wave takes the cosine. $\cos\theta_i = \frac{1}{\sqrt{3}}$

The x incident wave takes the sine. $\sin\theta_i = \frac{\sqrt{2}}{\sqrt{3}}$

$\theta_i = 54.7^o$

From Snell's law, $\sin\theta_t = \frac{\sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1}}\sin\theta_i$ $\theta_t = 35.3^o$

$\beta_o = 2.1$ rad/m $\beta_1 = 2.1\sqrt{2}$ rad/m

$\vec{H_r} = \frac{(-0.333)}{(377)(\sqrt{2})} e^{j\beta_o(\sqrt{2}x+z)/\sqrt{3}}\,(\vec{a}_x - \sqrt{2}\vec{a}_z)/\sqrt{3}$ A/m

$\vec{E_t} = \frac{(0.667)}{(\sqrt{2})} e^{-j\sqrt{2}\beta_o(-x+\sqrt{2}z)/\sqrt{3}}\,(\vec{a}_y)$ V/m

$\sqrt{\varepsilon_1}\sin\theta_i = \sqrt{\varepsilon_2}\sin\theta_t$

$\Gamma_{TE} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)} = -0.333$

$\tau_{TE} = 1 + \Gamma_{TE} = 0.667$

Single Interface: Oblique Incidence: TE

Incident Wave

$$\vec{E_i} = E_o\, e^{-j\beta_1(\cos\theta_i z + \sin\theta_i x)}\, \vec{a}_y$$

$$\vec{H_i} = \frac{E_o}{\eta_1} e^{-j\beta_1(\cos\theta_i z + \sin\theta_i x)}\,(-\cos\theta_i\, \vec{a}_x + \sin\theta_i\, \vec{a}_z)$$

Reflected Wave

$$\vec{E_r} = \Gamma_{TE} E_o\, e^{-j\beta_1(\sin\theta_r x - \cos\theta_r z)}\, \vec{a}_y$$

$$\vec{H_r} = \frac{\Gamma_{TE} E_o}{\eta_1} e^{-j\beta_1(\sin\theta_r x - \cos\theta_r z)}\,(\cos\theta_r\, \vec{a}_x + \sin\theta_r\, \vec{a}_z)$$

Transmitted Wave

$$\vec{E_t} = \tau_{TE} E_o\, e^{-j\beta_2(\cos\theta_t z + \sin\theta_t x)}\, \vec{a}_y$$

$$\vec{H_t} = \frac{\tau_{TE} E_o}{\eta_2} e^{-j\beta_2(\cos\theta_t z + \sin\theta_t x)}\,(-\cos\theta_t\, \vec{a}_x + \sin\theta_t\, \vec{a}_z)$$

$\theta_r = \theta_i$

$\beta_1 \sin\theta_i = \beta_2 \sin\theta_t$

$\sqrt{\varepsilon_{r1}}\sin\theta_i = \sqrt{\varepsilon_{r2}}\sin\theta_t$

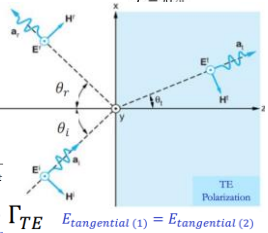
$$\tau_{TE} = \frac{2\,\eta_2\cos\theta_i}{\eta_2\cos\theta_i + \eta_1\cos\theta_t}$$

$$\tau_{TE} = 1 + \Gamma_{TE}$$

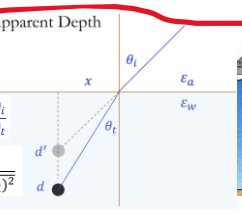
$R_{TE} = |\Gamma_{TE}|^2$

$T_{TE} = \frac{\eta_1\cos\theta_i}{\eta_2\cos\theta_i}|\tau_{TE}|^2$ $\Gamma_{TE} = \frac{\eta_2\cos\theta_t - \eta_1\cos\theta_i}{\eta_2\cos\theta_t + \eta_1\cos\theta_i} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)}$

$T_{TE} + R_{TE} = 1$



Apparent Depth

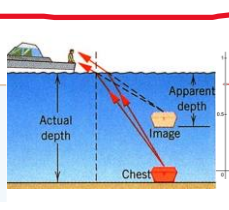


$d = x\cot\theta_t$ $\frac{d'}{d} = \frac{\cot\theta_i}{\cot\theta_t} = \frac{\cos\theta_i}{\cos\theta_t}\cdot\frac{\sin\theta_t}{\sin\theta_i}$

$d' = x\cot\theta_i$ $\frac{d'}{d} = \frac{\sqrt{\varepsilon_{ra}}}{\sqrt{\varepsilon_{rw}}}\cdot\frac{\cos\theta_i}{\cos\theta_t}$

$\sqrt{\varepsilon_{ra}}\sin\theta_i = \sqrt{\varepsilon_{rw}}\sin\theta_t$

$$\frac{d'}{d} = \frac{\cos\theta_i}{\sqrt{\varepsilon_{rw} - (\sin\theta_i)^2}}$$



The Optical Fiber.

The acceptance angle θ_a should correspond to the critical angle inside the fiber at the cladding-fiber interface. $\theta_i = \theta_c = 90 - \theta_b$

Snell's Law $\sqrt{\varepsilon_{ra}}\sin\theta_a = \sqrt{\varepsilon_{rf}}\sin\theta_b$

$\sin\theta_a = \frac{\sqrt{\varepsilon_{rf}}}{\sqrt{\varepsilon_{ra}}}\sin\theta_b = \frac{\sqrt{\varepsilon_{rf}}}{\sqrt{\varepsilon_{ra}}}\sin(90 - \theta_c)$

$\sin\theta_a = \frac{\sqrt{\varepsilon_{rf}}}{\sqrt{\varepsilon_{ra}}}\cos\theta_c = \frac{\sqrt{\varepsilon_{rf}}}{\sqrt{\varepsilon_{ra}}}\sqrt{1 - (\sin\theta_c)^2}$

Critical angle condition

$\sin\theta_c = \frac{\sqrt{\varepsilon_{rc}}}{\sqrt{\varepsilon_{rf}}}$ $\Rightarrow$ $\sin\theta_a = \frac{\sqrt{\varepsilon_{rf} - \varepsilon_{rc}}}{\sqrt{\varepsilon_{ra}}}$

$\varepsilon_{ra} = 1, \varepsilon_{rf} = 2.22, \varepsilon_{rc} = 2.19$

$\theta_c = 83.4^o$

$\theta_a = 9.9^o$

